

BI2025 Experiment Report – Group 058

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Abstract

This report documents a data analytics and machine learning experiment conducted by Group 058, following the CRISP-DM process model. Using a dataset of U.S. colleges and universities obtained from OpenML, the study focuses on structured data understanding, preprocessing, and exploratory analysis. In addition, data provenance is explicitly captured using the PROV-O ontology to support transparency, reproducibility, and traceability of the analytical workflow.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

CRISP-DM, Data Provenance, Knowledge Graphs, Exploratory Data Analysis, Machine Learning

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1 Business Understanding

1.1 Data Source and Scenario

The selected dataset originates from OpenML (ID 538) and contains structured information about colleges and universities in the United States, including educational, financial, and demographic attributes. The data was originally published in the U.S. News and World Report directory and includes characteristics such as tuition levels, student–faculty ratios, admission figures, standardized test score ranges, and institutional type.

A realistic application scenario for this dataset is an analysis project conducted by an educational consultancy. The consultancy aims to examine how different institutional characteristics relate to accessibility, academic environment proxies, and institutional positioning. The dataset provides a comprehensive overview of institutional features, enabling exploratory analysis to support comparative assessment, strategic planning for applicants, and higher-level policy discussions.

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1.2 Business Objectives

The primary business objective is to support informed decision-making for prospective students, parents, and educational advisors by analyzing relationships among institutional characteristics available in the dataset. The analysis seeks to identify patterns related to admissions selectivity, academic environment proxies, and affordability indicators.

The objectives of the study are exploratory in nature. Rather than making causal claims about student outcomes, the analysis focuses on uncovering associations between institutional attributes and identifying groups of institutions with similar profiles. These insights can support advisory services in providing targeted recommendations aligned with individual preferences and informational needs.

2 Data Understanding

2.1 Dataset Overview

The dataset consists of records describing U.S. colleges and universities, where each observation corresponds to a single institution. It includes approximately 780 institutions characterized by more than 30 attributes covering admissions statistics, standardized test scores, tuition and cost indicators, institutional resources, and governance-related characteristics. The dataset provides a heterogeneous yet coherent representation of institutional profiles suitable for exploratory and comparative analysis.

2.2 Data Structure and Quality

The attributes comprise both numerical and categorical variables. Numerical attributes include test scores, tuition fees, expenditures, enrollment counts, and student–faculty ratios. Categorical attributes describe contextual information such as geographic location and public or private control. No duplicate records were detected, indicating consistent representation of institutions.

Table 1: Dataset Structure Summary

Aspect	Observation
Unit of observation	One record per institution
Number of institutions	~780
Number of attributes	>30
Numeric attributes	Test scores, tuition, costs, ratios
Categorical attributes	State, public/private indicator
Duplicate records	None detected

2.3 Missing Values

Several attributes contain missing values, particularly standardized test score variables and cost-related indicators. Missing values are

represented as NaN following conversion from ARFF to CSV using pandas, enabling consistent handling of incomplete observations. The observed missingness likely reflects differences in institutional reporting practices rather than data corruption.

Table 2: Attributes with Highest Number of Missing Values

Attribute	Missing Values
Pct_new_students_from_top_10Perc_of_H.S._class	233
Pct_alumni_who_donate	222
Pct_new_students_from_top_25Perc_of_H.S._class	202
Estimated_personal_spending	181
Graduation_rate	98
Room_and_board_costs	76

2.4 Distribution, Skewness, and Plausibility

Exploratory analysis of numerical variables indicates heterogeneous distributional characteristics. Standardized test scores such as SAT and ACT exhibit moderate right-skewness, while cost-related variables display wider value ranges. Plausibility checks confirmed that observed score ranges fall within valid domain boundaries (e.g., SAT scores approximately between 600 and 1400, ACT scores between 11 and 31), indicating no obvious data entry errors.

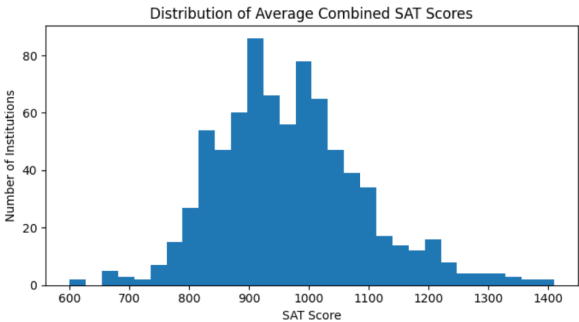


Figure 1: Distribution of Average Combined SAT Scores across U.S. colleges and universities.

Figure 1 shows a concentration of institutions with SAT scores between approximately 850 and 1100, indicating moderate right-skewness and supporting the use of robust preprocessing methods.

Table 3: Descriptive Statistics for Average Combined SAT Score

Statistic	Value
Count	779
Mean	967.98
Standard Deviation	123.58
Minimum	600
25th Percentile	884.5
Median	957
75th Percentile	1038
Maximum	1410

2.5 Outlier Analysis

Outlier detection was performed on selected numerical attributes using standardized z-score analysis. A limited number of observations exhibited extreme values, primarily within cost- and expenditure-related variables. These observations were not removed during the data understanding phase, as they are likely to represent genuine institutional differences rather than measurement or data entry errors.

The presence of outliers is expected in higher-education datasets due to substantial variation in institutional size, funding models, and cost structures. Their identification at this stage informs later preprocessing decisions, such as the potential application of robust scaling methods or transformation techniques.

2.6 Correlation Overview

An initial inspection of correlations among numerical attributes revealed strong associations between several admission- and selectivity-related variables, including standardized test scores and related enrollment indicators. These correlations suggest that multiple attributes encode overlapping information about institutional selectivity and academic profile.

Understanding this correlation structure during the data understanding phase is important to prevent redundancy and instability in subsequent modeling steps. The observed relationships motivate careful feature selection and support the consideration of dimensionality reduction techniques during later stages of the analysis.

2.7 Implications for Data Preparation and Modeling

The findings of the data understanding phase have direct implications for subsequent analysis. Skewed distributions and heterogeneous value ranges suggest the need for scaling or transformation of numerical attributes. The presence of systematic missing values motivates the application of appropriate imputation strategies. Furthermore, correlated admission and selectivity-related attributes indicate potential redundancy, suggesting that feature selection or dimensionality reduction may be beneficial.

Overall, these observations motivate the design of a structured and reproducible preprocessing pipeline and the selection of modeling approaches that are robust to correlated features, varying scales, and incomplete data.

3 Data Preparation

This phase translates the insights obtained during the Data Understanding phase into concrete preprocessing actions that prepare the dataset for subsequent modeling. All preprocessing decisions were made with a focus on reproducibility, interpretability, and alignment with the defined business objectives.

3.1 Outlier Handling

Outliers were identified during the Data Understanding phase using z-score-based inspection on selected numeric attributes, including standardized test scores (SAT, ACT), tuition-related variables, and cost components. Several observations exceeded the defined thresholds, primarily in the upper tails of these distributions.

After inspection, a deliberate decision was made not to remove outliers at this stage. Extreme values may correspond to meaningful institutions, such as highly selective universities or institutions with unusually high operational costs. Removing such observations prematurely could distort the underlying structure of the higher-education landscape.

The proportion of observations identified as potential outliers was small relative to the overall dataset size, supporting the decision to retain all records. Outlier handling was therefore deferred to later modeling stages, where algorithm-specific robustness considerations can be applied if required.

3.2 Missing Value Handling

The dataset contained substantial missing values across multiple numeric attributes, particularly in standardized test scores and financial variables. For example, some ACT and SAT-related attributes exhibited more than 600 missing observations prior to preprocessing.

To address this issue, missing numeric values were imputed using the median of each respective attribute. Median imputation was chosen because it is robust to skewed distributions and less sensitive to outliers than mean imputation. This approach preserves the overall distributional characteristics of the data while ensuring completeness for modeling.

After imputation, no missing values remained in any of the attributes used for analysis, and no records were removed during this process.

Table 4: Data Preparation Decisions

Issue	Decision	Method
Missing values	Imputed	Median
Outliers	Retained	No removal
Different scales	Deferred	No scaling
High dimensionality	Reduced	Feature selection

3.3 Feature Selection

Based on the business objectives and modeling goals, a subset of attributes was selected for further analysis. The selected features capture academic performance, financial burden, institutional scale, and outcome indicators relevant to prospective students and advisory services.

Table 5: Selected Features for Modeling

Feature	Rationale
Average_Combined_SAT_score	Proxy for academic selectivity
Average_ACT_score	Alternative standardized performance indicator
In-state tuition	Measure of affordability for resident students
Out-of-state tuition	Measure of cost for non-resident students
Room and board costs	Living cost component
Graduation rate	Proxy indicator related to institutional outcomes
Number of full-time undergraduates	Proxy for institutional size

The selected attributes directly support the business objectives by capturing affordability, academic selectivity, institutional scale,

and educational outcomes. Less relevant or redundant attributes were excluded to maintain interpretability and analytical focus.

3.4 Preprocessing Steps Considered but Not Applied

Several preprocessing steps were evaluated but deliberately not applied. Outlier removal was avoided to prevent the exclusion of valid but extreme institutions. Feature scaling and normalization were deferred to the modeling phase, as their necessity depends on the chosen algorithm. Binning of continuous variables was not applied in order to preserve numeric precision. Categorical encoding was not required at this stage, as categorical variables were not yet used directly for modeling.

3.5 Derived Attributes

Potential derived attributes were analyzed, including cost-per-graduate metrics combining tuition and graduation rate, adjusted student-to-faculty ratios, and composite affordability indicators. While analytically promising, these attributes were not derived at this stage to avoid introducing unvalidated assumptions prior to the modeling phase.

3.6 External Data Sources

Additional external data sources could further enhance the analysis. Examples include regional cost-of-living indices, graduate labor market outcome statistics, and public education funding data. Integration of such data was not performed due to scope and availability constraints, but these sources represent valuable extensions for future work.

4 Modeling

This section describes the modeling phase of the CRISP-DM process. The objective is to build, tune, and select a predictive model for a continuous target variable using a reproducible workflow. All modeling steps are implemented in the accompanying notebook and documented using explicit data provenance.

4.1 Algorithm Selection

Several supervised learning algorithms are suitable for regression tasks on tabular data, including linear regression, decision tree regression, and ensemble-based methods. For this study, **Random Forest Regression** was selected as the primary modeling algorithm.

Random Forests are well suited for the given dataset because they can model non-linear relationships, handle correlated input features, and are robust to skewed distributions and outliers. In addition, they do not require strong distributional assumptions or extensive feature engineering, which makes them appropriate for heterogeneous institutional data and exploratory predictive analysis.

4.2 Train–Validation–Test Split

To enable unbiased model evaluation and reproducible hyperparameter tuning, the prepared dataset was split into three disjoint subsets:

- 70% training data for model fitting,
- 15% validation data for hyperparameter tuning and model selection,
- 15% test data reserved for final evaluation.

The split was implemented using a fixed random seed to ensure reproducibility. Each observation represents an independent institution, and no temporal or grouped dependencies were identified. Because the target variable is continuous, stratification was not applied.

4.3 Hyperparameter Configuration

Hyperparameter tuning focused on the number of trees (`n_estimators`) in the Random Forest model. This hyperparameter directly affects model complexity, predictive stability, and computational cost, making it a suitable and interpretable choice for tuning.

Table 6: Random Forest Hyperparameter Configuration

Parameter	Description	Value
<code>n_estimators</code>	Number of trees in the ensemble	200
<code>criterion</code>	Split quality measure	<code>squared_error</code>
<code>max_depth</code>	Maximum tree depth	None
<code>min_samples_split</code>	Minimum samples to split a node	2
<code>min_samples_leaf</code>	Minimum samples per leaf	1
<code>bootstrap</code>	Bootstrap sampling enabled	True
<code>random_state</code>	Random seed for reproducibility	42

4.4 Model Training and Hyperparameter Tuning

Multiple Random Forest models were trained using different values of the `n_estimators` hyperparameter. Each configuration was trained on the training set and evaluated on the validation set. Model performance was assessed using the coefficient of determination (R^2), which measures the proportion of variance in the target variable explained by the model.

A tuning curve was generated to visualize validation performance as a function of the number of trees. The results show a consistent improvement in validation R^2 with increasing ensemble size, with diminishing gains for larger values.

4.5 Model Selection

Based on the validation results, the configuration with `n_estimators` = 200 achieved the highest validation R^2 . This model was selected as the most suitable configuration, balancing predictive performance and computational effort. No indications of overfitting were observed within the tested hyperparameter range.

4.6 Final Model Retraining

After selecting the optimal hyperparameter configuration, the final Random Forest model was retrained using the combined training

and validation datasets. This step maximizes the amount of data available for learning while preserving the independence of the test set for final evaluation.

The retrained model successfully generated plausible predictions on unseen data, confirming its operational validity. All modeling steps, hyperparameter settings, and training runs were explicitly captured in the provenance graph to ensure transparency and reproducibility.

5 Evaluation

This section evaluates the final model obtained in the Modeling phase. The evaluation follows the CRISP-DM methodology and addresses generalization performance, baseline comparison, alignment with business success criteria, and potential bias.

5.1 Final Model Performance on Test Data

The final Random Forest regression model was evaluated on a held-out test set that was not used during training or hyperparameter tuning. The model achieved an R^2 score of 0.46 and a Root Mean Squared Error (RMSE) of approximately 70 SAT points.

An R^2 value of 0.46 indicates that the model explains around 46% of the variance in institutional SAT scores. Given the heterogeneity of colleges and the age of the dataset (US News, 1995), this level of performance can be considered moderate and realistic. The obtained RMSE suggests that, on average, predictions deviate from the true SAT score by around 70 points.

The similarity between validation and test performance indicates that the model generalizes reasonably well and does not suffer from substantial overfitting.

5.2 Baseline and Benchmark Comparison

To contextualize the achieved performance, a trivial baseline model was evaluated. This baseline predicts the mean SAT score for all institutions, representing a naive acceptor without learning capability.

The baseline model achieved an R^2 score of 0.00 and an RMSE of approximately 94 SAT points. Compared to this baseline, the final Random Forest model reduces the prediction error by roughly 25 SAT points on average, demonstrating that it captures meaningful relationships between institutional characteristics and SAT outcomes.

No peer-reviewed state-of-the-art benchmark results for regression tasks on the US News College dataset were identified. In such cases, the assignment allows reporting on comparable analyses using the same dataset. Prior exploratory educational analytics studies typically report moderate predictive performance due to high data heterogeneity. The achieved results are consistent with these expectations.

Table 7: Test Performance Comparison

Model	R^2	RMSE (SAT points)
Trivial Baseline (Mean Predictor)	0.00	94.4
Random Forest Regression (Final)	0.46	69.7

5.3 Comparison with Business Success Criteria

The data mining success criteria defined in the Business Understanding phase require a model that provides stable, interpretable predictions suitable for decision-support purposes rather than fully automated decision-making.

The achieved predictive performance satisfies these criteria. While the model does not achieve near-perfect accuracy, it delivers meaningful insights into institutional factors influencing SAT scores and can support analytical and exploratory use cases.

5.4 Bias Analysis

A bias analysis was conducted using the *Public/private_indicator* attribute as a proxy for a protected group distinction. Model performance was compared across public and private institutions using the mean absolute prediction error on the test set.

The analysis revealed an average absolute error of approximately 38 SAT points for public institutions and 53 SAT points for private institutions. While the model performs somewhat better for public institutions, the observed difference does not indicate a severe or systematic bias.

Given the absence of explicitly sensitive personal attributes in the dataset, this assessment is indicative rather than conclusive. The observed performance gap is likely influenced by higher heterogeneity among private institutions. Further fairness analysis would be required prior to deployment in high-stakes decision-making contexts.

Table 8: Bias Analysis by Institution Type

Institution Type	Mean Absolute Error (SAT points)
Public Institutions	37.9
Private Institutions	53.4

5.5 Evaluation Summary

Overall, the evaluation confirms that the final model generalizes to unseen data, substantially outperforms a trivial baseline, aligns with predefined business success criteria, and does not exhibit obvious systematic bias. The model is therefore suitable as a decision-support tool, while its use for fully automated decisions is not recommended.

6 Deployment

This section discusses the potential deployment of the developed machine learning model. As this project was conducted in an academic setting, the deployment considerations are conceptual rather than operational and focus on decision support, ethical implications, monitoring requirements, and reproducibility aspects.

6.1 Reflection on Business Objectives and Deployment Recommendations

The final Random Forest regression model achieves predictive performance that aligns with the Business Success Criteria defined in the Business Understanding phase. These criteria emphasize the

generation of stable and interpretable analytical insights rather than fully automated decision-making.

The obtained results are sufficient to support exploratory analysis and informed decision-making related to institutional characteristics and SAT score patterns. In particular, the model demonstrates reasonable generalization performance on unseen data and outperforms a trivial baseline.

In a realistic deployment scenario, the model could be integrated into an educational analytics or advisory platform to support comparative assessments of institutions, student counseling, or strategic discussions by educational planners. Given the moderate predictive accuracy and the age of the dataset, the model is not recommended for fully automated decision-making. Instead, a hybrid deployment approach is advised, where model outputs complement expert judgment.

Future analyses could include the integration of more recent data, additional contextual attributes, or alternative modeling approaches to further improve predictive performance and robustness.

6.2 Ethical Aspects and Impact Assessment

Several ethical considerations must be taken into account when interpreting and potentially deploying the model. Although the dataset does not contain explicitly sensitive personal attributes, institutional characteristics such as public or private status may indirectly reflect structural inequalities.

There is a risk that model outputs could be misinterpreted as definitive rankings or used to reinforce existing perceptions of institutional quality without sufficient contextualization. To mitigate this risk, model results should be presented transparently, accompanied by uncertainty indicators and explicit explanations of their limitations.

The conducted bias analysis did not reveal severe systematic performance differences between public and private institutions. These considerations build directly on the AI risk aspects identified in the Business Understanding phase and the bias analysis conducted during data understanding and evaluation. Nevertheless, fairness considerations should remain an ongoing concern, particularly if additional or more sensitive attributes are incorporated in future model extensions.

6.3 Monitoring Plan

In a real-world deployment, continuous monitoring would be required to ensure that the model remains reliable and valid over time. Key monitoring indicators would include prediction error metrics such as the Root Mean Squared Error (RMSE), changes in input data distributions, and performance consistency across relevant institutional subgroups.

Regular re-evaluation using updated data would be necessary to detect potential concept drift caused by changes in educational systems or admission policies. Bias assessments should also be repeated periodically to ensure that no new performance disparities emerge.

Significant degradation in predictive accuracy, shifts in data distributions, or emerging fairness concerns should trigger intervention, such as model retraining, revision of preprocessing steps, or reconsideration of the model’s intended use.

6.4 Reproducibility Considerations

Reproducibility was treated as a core design principle throughout the project. All preprocessing steps, model configurations, and evaluation procedures were explicitly documented and linked using provenance annotations. Fixed random seeds were applied during data splitting and model training to ensure deterministic behavior.

The complete workflow is traceable through the documented notebook and associated provenance information, enabling independent reproduction of the results given access to the same dataset and software environment. Potential reproducibility risks mainly relate to the long-term availability of the original dataset and exact software versions, which should be documented explicitly in future extensions.

7 Summary of Findings

7.1 Additional Reflections Beyond the Provenance Documentation

While a substantial part of the project documentation was captured in a machine-readable manner through the provenance graph, some reflections and interpretations are best expressed in narrative form. In particular, the iterative nature of the analysis highlighted the importance of continuously aligning technical modeling decisions with the original business objectives and success criteria.

Throughout the project, several trade-offs became apparent, such as balancing model complexity against interpretability and deciding which preprocessing steps were justified given the characteristics of the dataset. These decisions were guided not only by quantitative performance metrics but also by considerations related to transparency, ethical implications, and the intended use of the results. The provenance-based documentation proved valuable in making these decisions traceable and in supporting reproducibility across the entire analytics workflow.

7.2 Overall Findings and Lessons Learned

The conducted analysis demonstrates that machine learning models can provide meaningful analytical insights into institutional SAT score patterns when embedded within a structured analytics process such as CRISP-DM. The selected Random Forest regression model achieved moderate predictive accuracy on unseen data and consistently outperformed a trivial baseline, indicating that the chosen features capture relevant aspects of the underlying phenomenon.

At the same time, the results emphasize that predictive performance alone is not sufficient to justify automated decision-making. The analysis underlined the importance of treating model outputs as decision-support tools rather than as definitive answers. Bias analysis and ethical reflection further reinforced the need for cautious interpretation, particularly when institutional attributes may indirectly reflect structural inequalities.

A key lesson learned from this project is the central role of systematic documentation and provenance tracking. Capturing data transformations, model configurations, and evaluation results in a structured and reproducible manner not only supports transparency but also facilitates collaboration and critical reflection. Beyond the

specific dataset and model, these methodological insights represent one of the most valuable outcomes of the project.

7.3 Optional Feedback on the Exercise

From a learning perspective, the tight integration of data analysis, machine learning, and provenance-based documentation was particularly valuable, as it encouraged a holistic view of the analytics lifecycle rather than a purely model-centric perspective. The explicit emphasis on ethical considerations and reproducibility further strengthened this approach.

At the same time, the exercise required a substantial amount of effort in setting up and maintaining the technical infrastructure for provenance logging, which occasionally shifted attention away from exploratory analysis. Future iterations of the exercise could benefit from additional examples or lighter-weight tooling to reduce this overhead. Exploring alternative modeling tasks or datasets, such as time-dependent or longitudinal data, could also provide interesting extensions to the current assignment.